Umbrella framework

According to Mathew Leibold, the study of metacommunities brings together many seemingly disparate branches of ecology and apparently conflicting theories. It may make it easier, for example, to resolve the century-old debate between the “deterministic,” niche-based theory of community ecology, in which species diversity is determined by each species’ ecological niche, and “stochastic” (random) theory, which emphasizes the importance of chance colonization and ecological drift (random fluctuations in population sizes).

Metacommunity theory provides an umbrella framework for seeing how deterministic and stochastic processes can interact to form natural communities. It allows ecologists to state that patterns of biodiversity are determined both by local biological features, such as the balance of sun and shade in rock pools or variations in water quality in streams, and by regional stochastic processes, such as the spread of a species by freak storms or a die-off due to an epidemic. It also acknowledges that regional changes can be caused by the combined effect of local ones.